Figure 2. Cross-substrate residence-pathology grammar across eight substrates
one principle, one observable $R(s, c) = \tau_R(s) \times \pi_c$, eight substrate-specific realizations

Substrate	State (s)	Compartment (c)	τ_R observable	π_c observable	Return-path machinery	Canonical pathology under residence failure
§8.1 Pharmacology	drug-target binding state	tissue compartment	1/k_off (residence time)	tissue-occupancy probability	drug clearance + tissue redistribution	long-residence toxicity
§8.2 Cancer phospho-reg	phospho-substrate state	intracellular localization	phospho-state dwell time	compartment occupancy	PP2A / GSK3β phosphatases	constitutive activation
§8.3 Autonomic physiology	sympathetic vs parasympathetic tone	baseline vs perturbation regime	recovery τ after stressor	regime occupancy	vagal-brake / baroreflex	depression, CV disease
§8.4 Exercise physiology	metabolic boundary state	muscle / cardiac / hepatic / neuroendocrine	boundary dwell time	compartment over-occupancy	endogenous cutoffs (PCr/H+/sympatho-exhaustion)	operator-sustained cardiomyopathy (HAARLEM)
§8.5 Brain network dynamics	metastable brain state (HMM)	regional occupancy	state dwell time	network occupancy probability	network-dynamical switching (CEN ↔ DMN)	depression, anxiety, rumination
§8.6.1 Memory phenomenology	retrieval-coupled state	present-axis vs past-axis grounding	retrieval coupling time	compartment occupancy	reconsolidation window	PTSD, regret, nostalgia, grief
§8.6.2 Public-epistemic	interpretive frame	info-source cluster	frame dwell time	cluster segregation index	cross-cutting exposure + fact-checking	echo chambers, misinformation persistence
§8.7 AI / data substrate	model-distribution mode	training-data regime	mode dwell under iter. inference	regime segregation	real-data circulation + human cognition	model collapse, epistemic degradation

Each row instantiates the residence-pathology principle at a substrate developed independently in its own literature. The framework's contribution is the explicit composition rule, not new substrate-level claims at any one row.